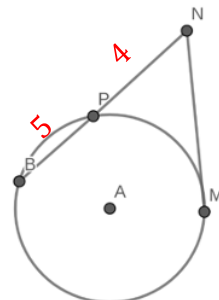


Geometry
Unit 12
Lesson 3 Practice

Name _____
Date _____
Period _____

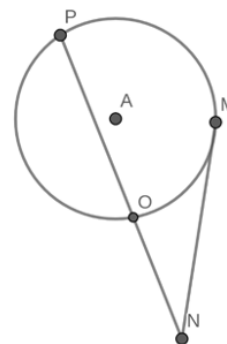
1. Circle A is shown with points M, B and P on the circle. Secant $\overline{BN}$ and tangent $\overline{MN}$ intersect at point N outside the circle. $PB = 5$ and $PN = 4$. What is the length of $\overline{MN}$?

$$\begin{aligned}PN(BP + PN) &= MN^2 \\4(5 + 4) &= MN^2 \\4(9) &= MN^2 \\\sqrt{36} &= \sqrt{MN^2} \\MN &= 6\end{aligned}$$



2. Circle A is shown with points M, O and P on the circle. Secant $\overline{PN}$ and tangent $\overline{MN}$ intersect at point N outside the circle. $PO = (3x + 3)$, $ON = 10$ and $MN = 20$. Find the value of x .

$$\begin{aligned}MN^2 &= 10(10 + 3x + 3) && \rightarrow 270 = 30x \\20^2 &= 130 + 30x && 9 = x \\400 &= 130 + 30x && x = 9\end{aligned}$$

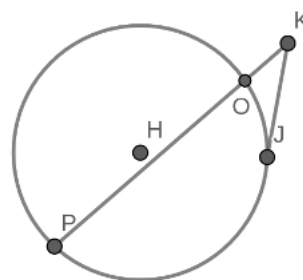


3. What is the length of $\overline{PN}$?

$$\begin{aligned}PN &= 13 + 3(9) \\&= 40 \\PN &= 40\end{aligned}$$

4. Circle H is shown with points J, O and P on the circle. Secant $\overline{KP}$ and tangent $\overline{KJ}$ intersect at point K outside the circle. $PO = 112$, $OK = 14$ and $KJ = x$. Find the value of x .

$$\begin{aligned}OK(PO + OK) &= KJ^2 \\14(112 + 14) &= x^2 \\14(126) &= x^2 \\\sqrt{1764} &= \sqrt{x^2} \\x &= 42\end{aligned}$$

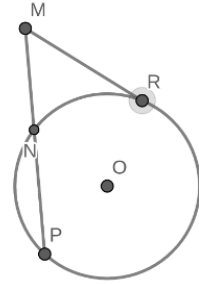


5. Using the same image as question 4, circle H is shown with points J, O and P on the circle. Secant $\overline{KP}$ and tangent $\overline{KJ}$ intersect at point K outside the circle. $PO = 16 + x$, $OK = 2$ and $KJ = 8$. Find the length of $\overline{KP}$.

$$\begin{aligned}2(16 + x + 2) &= 8^2 && KP = PO + OK \\2(18 + x) &= 64 && 16 + 14 + 2 \\36 + 2x &= 64 && KP = 32 \\2x &= 28 \\x &= 14\end{aligned}$$

6. Circle O is shown with points N, R and P on the circle. Secant $\overline{MP}$ and tangent $\overline{MR}$ intersect at point M outside the circle. $PN = 20$, $MN = 16$ and $MR = 4x$. What is the value of x ?

$$\begin{aligned} MN(PN + MN) &= MR^2 \\ 16(20 + 16) &= (4x)^2 \\ 576 &= 16x^2 \\ 36 &= x^2 \\ x &= 6 \end{aligned}$$

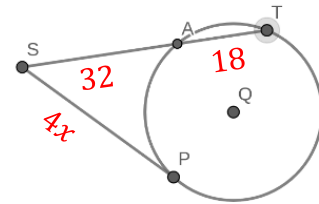


7. Using the same image as question 6, Circle O is shown with points N, R and P on the circle. Secant $\overline{MP}$ and tangent $\overline{MR}$ intersect at point M outside the circle. $PN = x + 10$, $MN = 15$ and $MR = 30$. What is the value of x ?

$$\begin{aligned} MN(PN + MN) &= MR^2 \\ 15(x + 10 + 15) &= 30^2 \\ 15x + 375 &= 900 \end{aligned} \quad \begin{aligned} &\rightarrow 15x = 525 \\ &x = 35 \end{aligned}$$

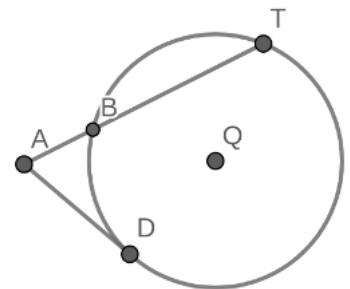
8. Circle Q is shown with points A, T and P on the circle. Secant $\overline{ST}$ and tangent $\overline{SP}$ intersect at point S outside the circle. $SP = 4x$, $AT = 18$ and $AS = 32$. What is the value of x ?

$$\begin{aligned} PS^2 &= AS(AS + TA) \\ (4x)^2 &= 32(18 + 32) \\ 16x^2 &= 1600 \\ x^2 &= 100 \\ x &= 10 \end{aligned}$$



9. Daniella and Taylor are working on finding the length of $\overline{AD}$. Circle Q is shown with points B, T and D on the circle. Secant $\overline{AT}$ and tangent $\overline{AD}$ intersect at point A outside the circle. $BT = 12$, $AB = 10$ and $AD = 2x$. Their partial work solving for x is shown below. Find the value of x for each of their work. Round to the nearest hundredth. Whose work do you agree with? **Daniella**

Daniella's work	Taylor's work
$(2x)^2 = 10(10 + 12)$ $4x^2 = 10(22)$ $x^2 = 55$ $x = 7.42$	$(2x)^2 = 12(10 + 12)$ $4x^2 = 12(22)$ $x^2 = 132$



10. Using the image in Question 9, what is the length of $\overline{AD}$?

$$\begin{aligned} AD &= 2x \\ &= 2 \cdot 7.42 \\ AD &= 14.84 \end{aligned}$$